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Roll No. 311241ue39

324555(24)

B. E. (Fifth Semester) Examination, Nov.-Dec. 2016

(New Scheme)

(Elect. Engg. Branch)

CONTROL SYSTEM ENGINEERING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: All questions are compulsory. Part (a) is compulsory and carries 2 marks. Out of (b), (c), (d), solve any two and carries 7 marks each.*

1. (a) Define control system.

(b) Write differential equations governing the mechanical system shown below and obtain the transfer function $X(S) / F(S)$.

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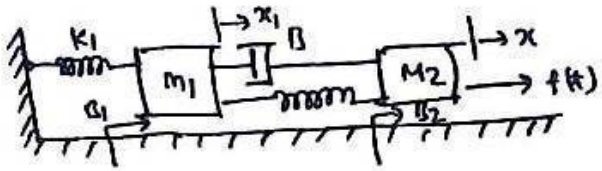
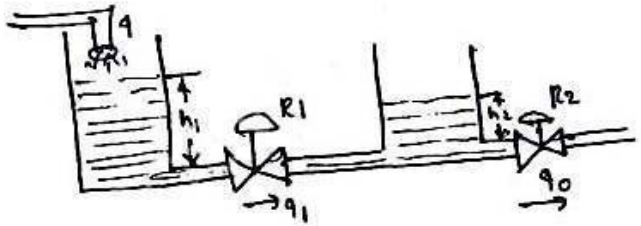


(c) Develop a mathematical model for the two coupled tank system. Derive the transfer function $H_2(J) / Q(S)$.



(d) Discuss the followings :

(i) Dead-time

(ii) Synchro

(iii) Linearity

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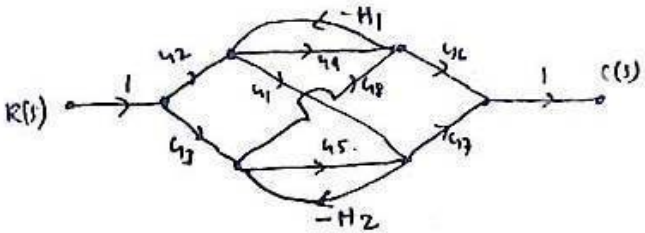
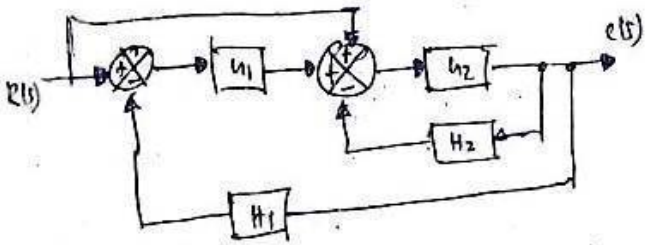
2. (a) What is a pneumatic actuator?

(b) Find the overall gain of the system for the given SFH.



(c) Derive the transfer function of armature controlled DC servomotor.

(d) Determine the closed loop transfer function using block diagramation method.



3. (a) Define stability and its types.

(b) Sketch the root locus of the system whose open loop transfer function is $G(s) = K/\{s(s+2)(s+4)\}$. find

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the value of K so that damping ratio of closed loop system is 0·5.

(c) The characteristics polynomial of a system is,

$$s^7 + 5s^6 + 9s^5 + 9s^4 + 4s^3 + 20s^2 + 36s + 36 = 0$$

Determine the location of room and stability of the system using Routh Hurwitz criterion.

(d) Determine the time domain response of second order undamped and critically dampad system for unit step input.

4. (a) Define gain margin and phase margin.

(b) Sketch the polar plot for $G(s) = \dfrac{1}{s(1+s)(1+2s)}$

and determine the gain margin and phase margin.

(c) Sketch the bode plot for the following transfer function

$$G(s) = \frac{75(1+0{\cdot}25)}{s(s^2+16s+100)}$$

(d) Construct a nyquist plot for a feedback control system whose open loop transfer function if given by

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$$G(s)H(s)=\frac{5}{s(1-S)}.$$

5. (a) Define compensation.

(b) Explain the procedure of Lag compensator design with not locus approach.

(c) Illustrate for procedure for phase lead compensating network using Bode plot.

(d) Consider a lag-lead compensator defined by

$$G_C(s)=K_C\frac{\left(s+\frac{1}{T_1}\right)\left(s+\frac{1}{T_2}\right)}{\left(S+\frac{\beta}{T_1}\right)\left(S+\frac{1}{\beta T_2}\right)}$$

Show that at frequency $w_1=\sqrt{\frac{1}{T_1T_2}}$, the phase angle of $G_C(jw)$ becomes zero.

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